



STEM CELL REGENERATION



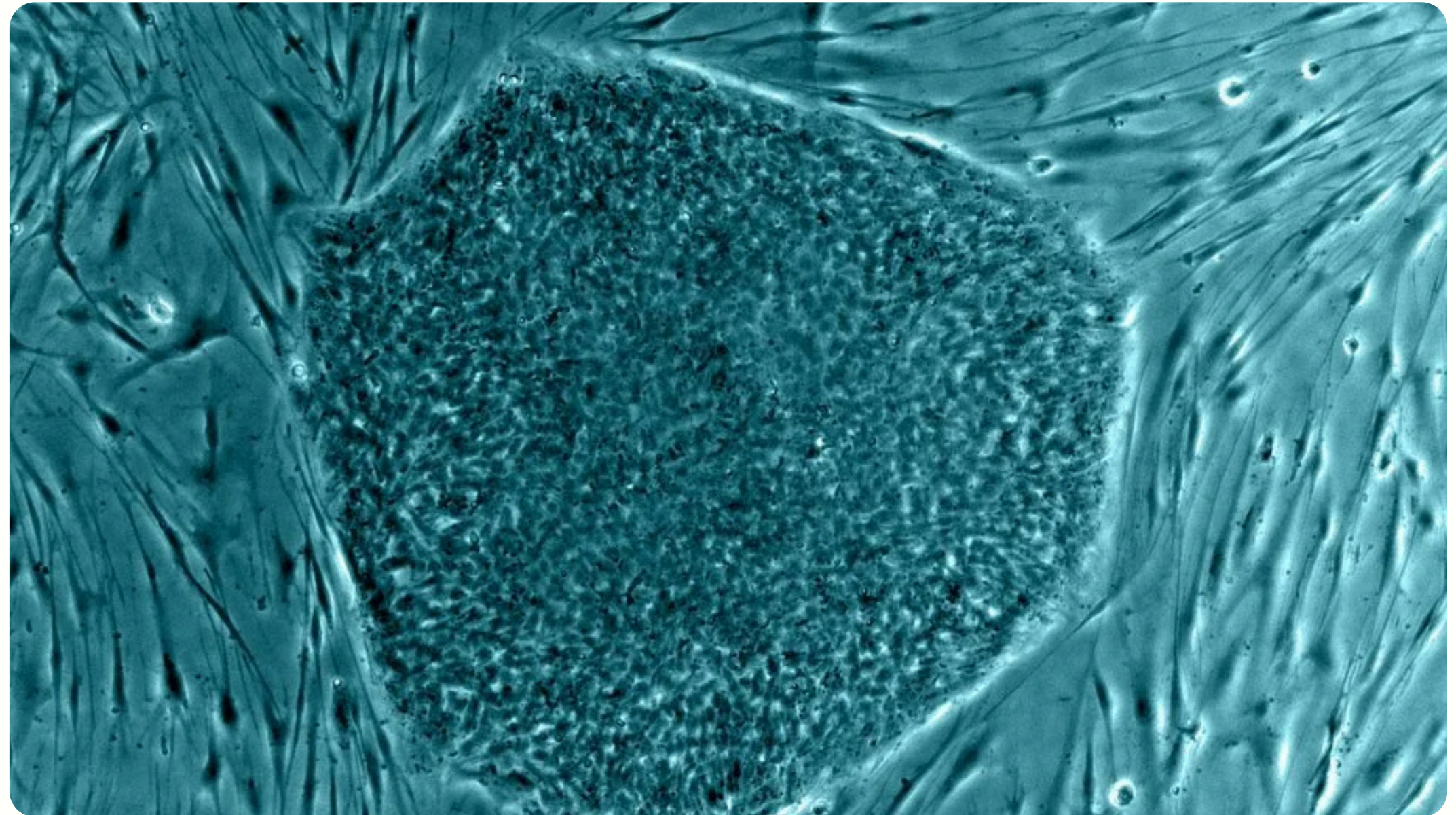
*** By Arya Asati (202501070064) ***
From ENTC ET1 E13





Introduction

- * What are stem cell ?
- * Unique Property : Self-renewal + Differentiation
- * Why important ? (healing, therapy, research)



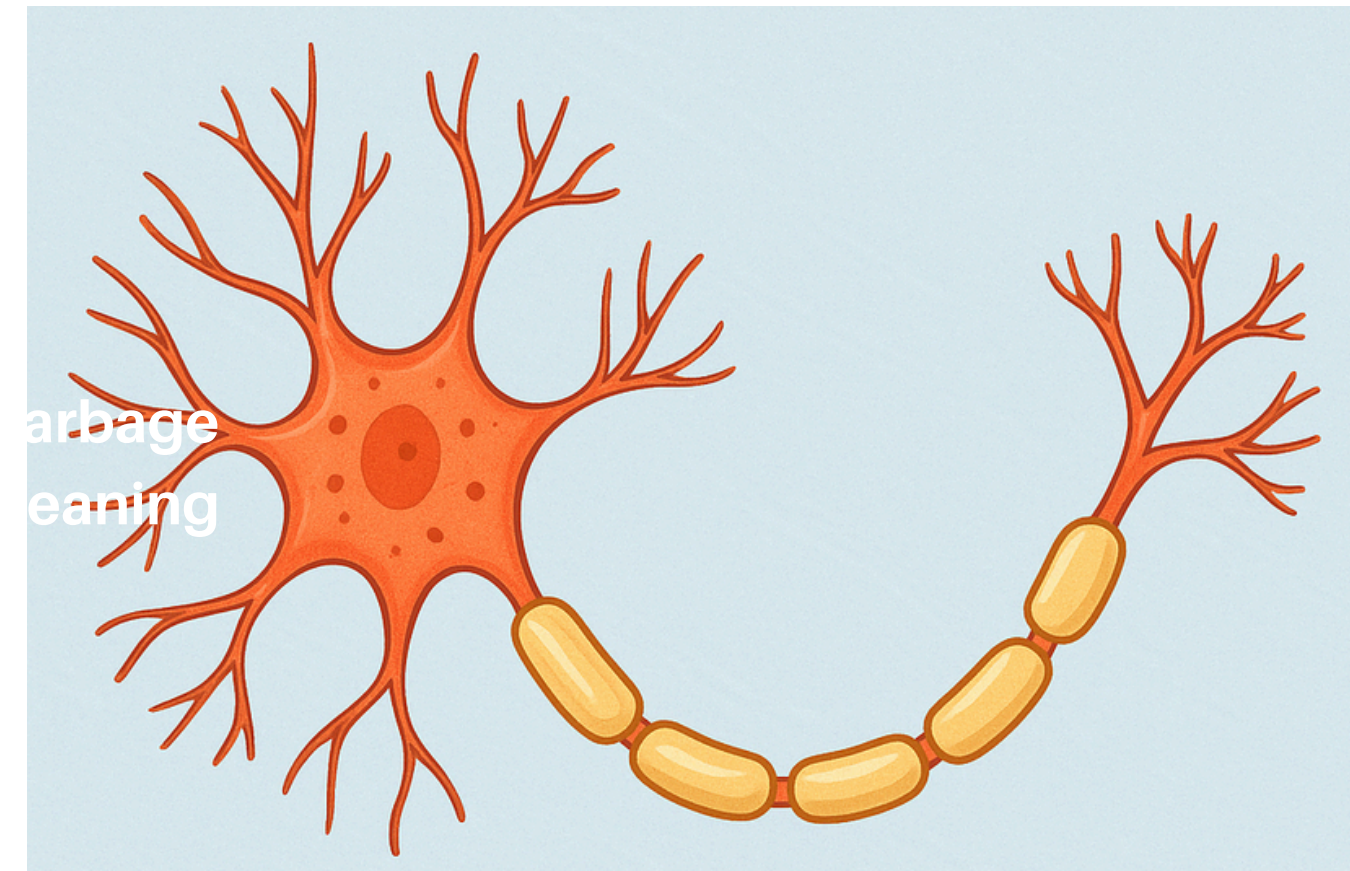
BACKGROUND :

- * Discovered in 1960s (bone marrow studies)**
- * Types: Embryonic, Adult, Induced Pluripotent (iPSCs)**
- * Key scientists (Ernst McCulloch & James Till)**



Basic Concept :

- * Stem cells can turn into specialized cells (e.g., nerve, muscle, blood)
- * Regenerate tissues after injury.
- * Natural in the body vs. Lab-grown.



Mechanism of Regeneration

- * Activation of dormant stem cells.**
- * Cell division----Differentiation--Tissue repair.**
- * Role of growth factors and signaling pathways, (notch,Wnt,Hedgehog).**



Types of Stem Cells :



*** Embryonic Stem Cells (pluripotent, can form any cell.**

*** Adult Stem Cells (multipotent, limited capacity).**

*** Induced Pluripotent Stem Cells (iPSCs)-lab-made, Nobel Prize 2012.**



Literature Survey :

- * Stem cell therapy in leukemia (bone marrow transplant).
- * Heart regeneration studies (Harvard, Mayo, Clinic).
- * Nerve injury repair (spinal cord research).



planting

conservation

cleaning

Recent Developments :



- * 3D organoids (mini organs in lab).
- * CRISPR gene editing with stem cells.
- * stem cell banking.
- * FDA-approved therapies in 2024-2025.

Case Study :



EX; Stem cell therapy in
parkinson's disease

Results : improvement in
motor function

Japan's clinical trials for
iPSCs



Applications :



Regenerative medicine
(heart,liver,spinal cord).



Treating blood disorders
(leukemia,thalassemia).



Organ transplantation alternative



Cosmetic/skin regeneration (burn patients).



Importance in daily life :

- ✱ **Potential to cure currently untreatable disease**
- ✱ **hope for aging-related tissue loss**
- ✱ **Biotechnology and pharmaceutical advancements**



Challenges/Limitations :



- * Ethical issues (embryonic stem cells).**
- * Risk of tumors (uncontrolled cell growth).**
 - * Expensive therapies.**
- * Limited availability of safe clinical trials.**

Future Scope :



- * Growing full organs in lab.**
- * Personalized medicine with patients own stem cells.**
- * Combining AI + Stem cell research.**
- * Wider global availability.**

Summary/Conclusion :



- * stem cells = foundation of regenerative biology.**
 - * Great promise in treating disease, but challenges remain.**
- * A game changer for the future of medicine.**

References :



- * Research papers**
- * Textbooks**
- * WHO**
- * PubMed articles**



THANK YOU!

